REGISTERS

SEQUENTIAL LOGIC CIRCUITS

prepared by:

Gyro A. Madrona

Electronics Engineer

TOPIC OUTLINE

Registers

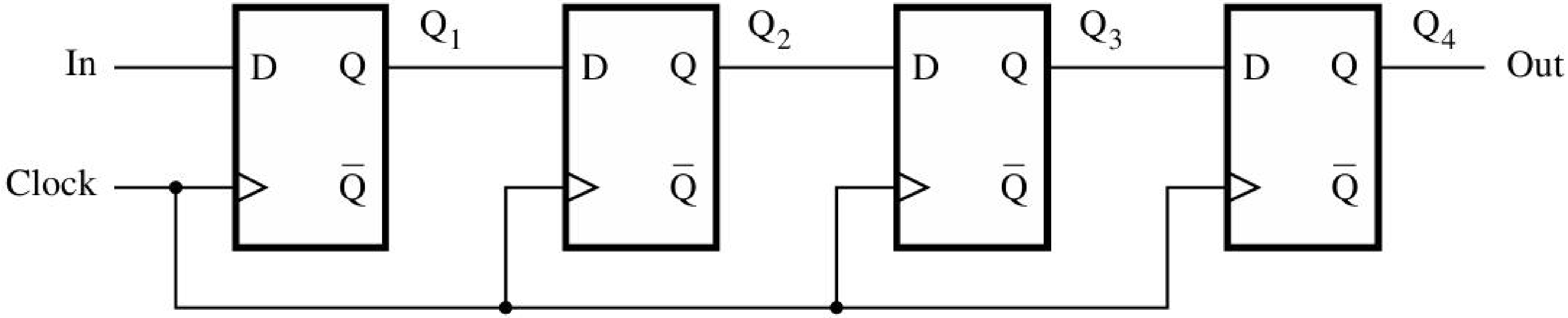
- **Shift Register**
- **Parallel-Access Shift Register**

REGISTERS

REGISTER

Flip-flop stores one bit of information.

Register store ***n* bits** of information.



A simple shift-register

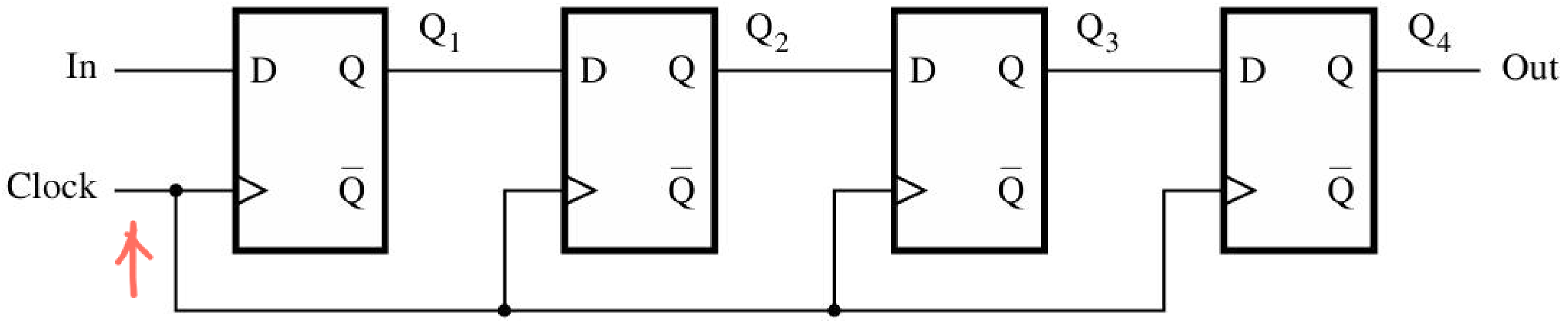
4-bit register = 4 flip-flops

The term register is merely a convenience for referring to n-bit structures consisting of flip-flops.

SHIFT REGISTER

A register that provides the ability to shift its contents is called a **shift register**.

```
sample sequence: 0 0 0 1 1 1 0 1
```



A simple shift-register

	In	Q_1	Q_2	Q_3	Q_4 = Out
t_0	1	0	0	0	0
t_1	0	1	0	0	0
t_2	1	0	1	0	0
t_3	1	1	0	1	0
t_4	1	1	1	0	1
t_5	0	1	1	1	0
t_6	0	0	1	1	1
t_7	0	0	0	1	1

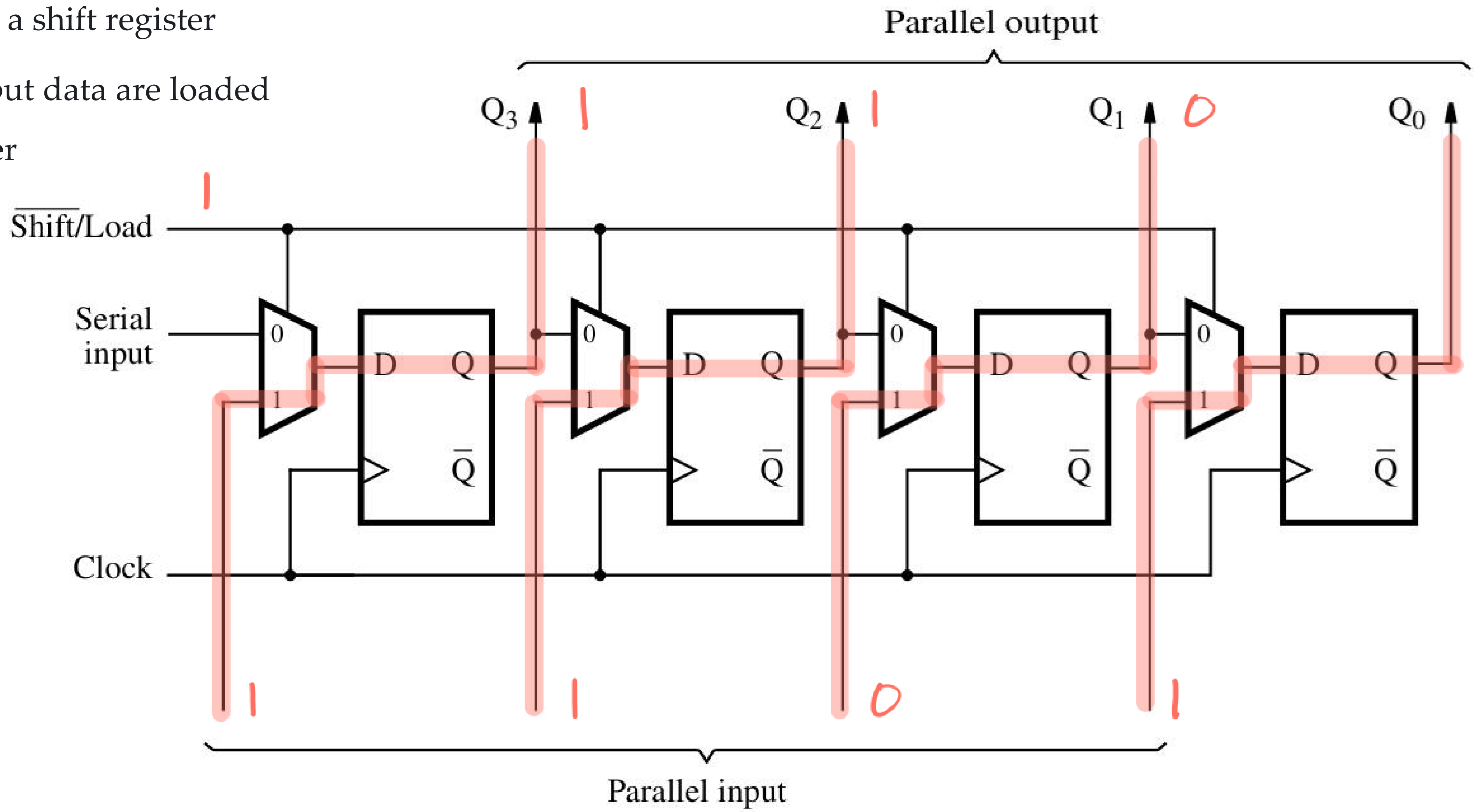
Shifted 1-bit every ↑

PARALLEL-ACCESS SHIFT REGISTER

The control signal $\overline{\textbf{Shift}}/\textbf{Load}$ is used to select the mode of operation.

0 – operates as a shift register

1 – parallel input data are loaded into the register

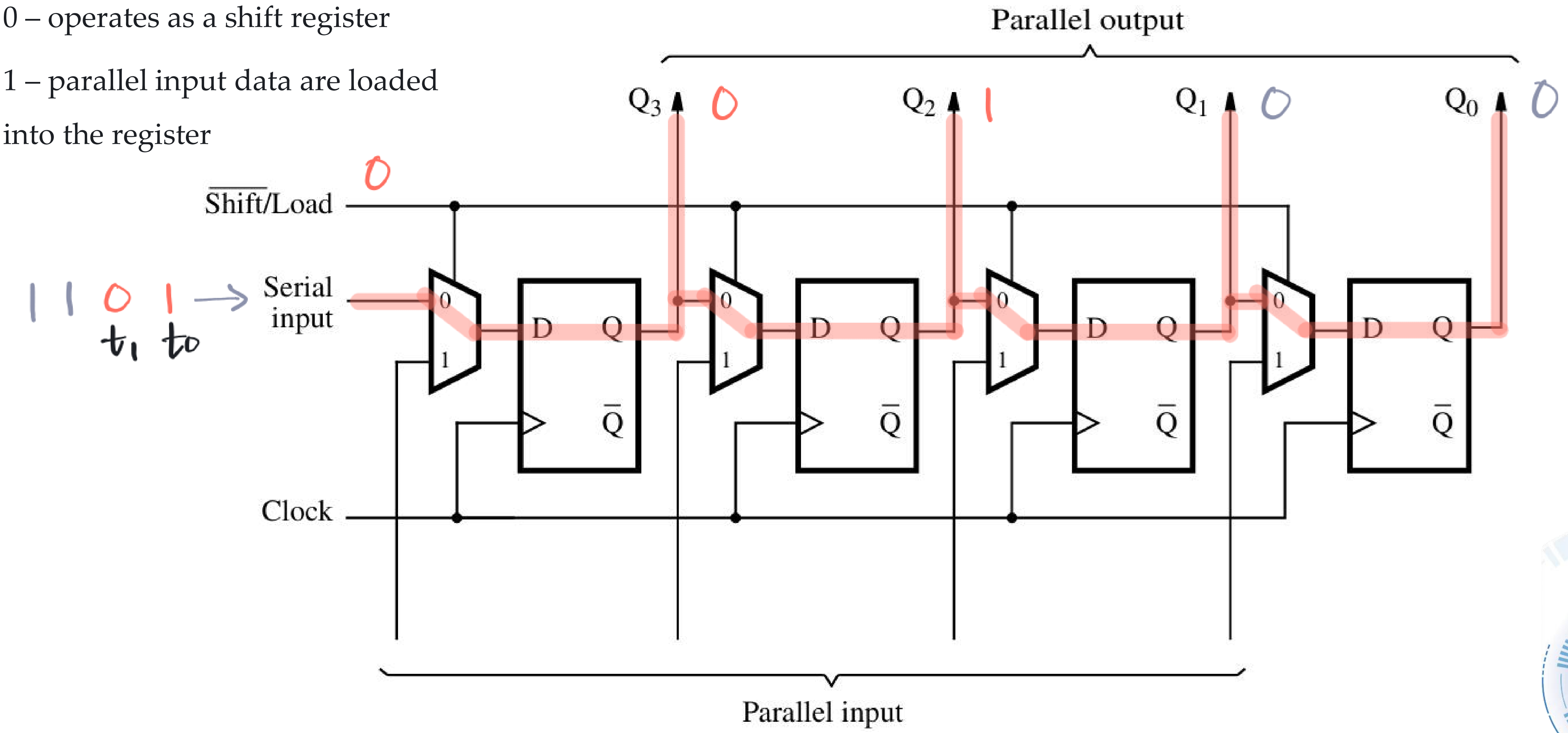


PARALLEL-ACCESS SHIFT REGISTER

The control signal $\overline{\textbf{Shift}}/\textbf{Load}$ is used to select the mode of operation.

0 – operates as a shift register

1 – parallel input data are loaded into the register



LABORATORY